

Amirpasha Hedayat

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RESEARCH INTERESTS

My research focuses on scientific machine learning and AI for science, with an emphasis on reduced-order models, operator learning, and neural approaches for modeling complex dynamical systems. Beyond specific applications, I'm interested in how AI can serve as a general-purpose modeling tool, capable of adapting across scientific domains and levels of abstraction.

EDUCATION

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|------------|---|----------------------|
| PhD | University of Michigan , Aerospace Engineering & Scientific Computing | Aug 2023 – Present |
| | <ul style="list-style-type: none"> • Also affiliated with Michigan Institute for Computational Discovery and Engineering (MICDE) • Advisor: Prof. Karthik Duraisamy | |
| MSc | University of British Columbia , Mechanical Engineering | Aug 2020 – Nov 2022 |
| | <ul style="list-style-type: none"> • GPA: 4.0/4.0 (92% local) • Advisor: Prof. Carl Ollivier-Gooch • Thesis: "Detecting Missing Flow Separation and Predicting Error in Drag using Supervised Machine Learning" [Link ↗] | |
| BS | Amirkabir University of Technology , Aerospace Engineering | Sept 2015 – Oct 2019 |
| | <ul style="list-style-type: none"> • GPA: 4.0/4.0 (19.0/20.0 local) • Advisor: Prof. Sahar Noori • Thesis: "Estimation of Unknown Heat Flux in a 2-D Heat-Sink using Levenberg-Marquardt Inverse Algorithm" | |

EXPERIENCE

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| University of Michigan , Graduate Research Assistant | Ann Arbor, MI, USA
Aug 2023 – Present |
| <ul style="list-style-type: none"> • Focus: Reduced-order Models, Scientific Machine Learning, AI for Science • Main Tools: Python | |
| University of British Columbia , Graduate Research Assistant | Vancouver, BC, Canada
Aug 2020 – Nov 2022 |
| <ul style="list-style-type: none"> • Focus: Dimensionality Reduction, Machine learning, Error Analysis, CFD • Main Tools: Python, C++, Matlab | |
| Amirkabir University of Technology , Undergraduate Research Assistant | Tehran, Iran
Sep 2017 – Oct 2019 |
| <ul style="list-style-type: none"> • Focus: Numerical Simulation, Inverse Problems, Heat Transfer, CFD • Main Tools: C++, Matlab, Ansys Fluent, Pointwise | |
| Sholeh Sanat Manufacturing and Engineering Company , Undergraduate Intern | Tehran, Iran
Jun 2018 – Aug 2018 |
| <ul style="list-style-type: none"> • Focus: Design and Modification of a Heat Exchanger • Main Tools: Autodesk Inventor, Ansys Fluent | |

PUBLICATIONS

- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Under-Resolved Flow Physics Using Supervised Machine Learning," *AIAA Journal*, June 2023. doi: <https://doi.org/10.2514/1.J062884> [↗](#)
- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Missing Flow Separation using Supervised Machine Learning," *AIAA SCITECH 2023 Forum*, National Harbor, MD, January 2023. doi: <https://doi.org/10.2514/6.2023-1201> [↗](#)

TALKS & POSTERS

- “Non-intrusive Adaptive Reduced-Order Models for Predictive Modeling of Complex Systems,” *18th U.S. National Congress on Computational Mechanics (USNCCM18)*, Chicago, IL, July 2025.
- “Linear and Non-linear Reduced Dimensional Manifolds for Global Weather Predictions,” *Model Reduction and Surrogate Modeling (MORe2024)*, San Diego, CA, September 2024.
- “Detecting Missing Flow Separation using Supervised Machine Learning,” *AIAA SCITECH 2023 Forum*, National Harbor, MD, January 2023.

SKILLS

Programming Languages: Python, Julia, C/C++, Matlab/Octave, SQL/MySQL, HTML/CSS

AI/ML Frameworks: Pytorch, Tensorflow/Keras, Scikit-learn

Development Tools: Git/GitHub, Autotools/Make, Shell Scripting

Operating Systems: Extensive experience with MacOS, Linux (Ubuntu), and Microsoft Windows

Spoken Languages: English (proficient), Persian (native)

AWARDS & HONORS

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| • Graduate Student Academic Achievement Award - UBC | 2022 |
| • Natural Sciences and Engineering Research Council of Canada (NSERC) Graduate Fellowship | 2020 - 2022 |
| • International Tuition Award - UBC | 2020 - 2022 |
| • Faculty of Applied Science Graduate Award - UBC | 2020 - 2022 |
| • The Distinguished Aerospace Engineering BSc Graduate - AUT | 2019 |
| • Graduated as the Top Student in Undergraduate Studies (out of ~150) - AUT | 2019 |
| • Iran's National Elites Foundation (INEF) Scholarship | 2018 - 2019 |
| • Department of Aerospace Engineering Academic Excellence Award - AUT | 2016 - 2019 |